

Bridging the Expressiveness Gap in Cross-Cultural Co-Creation through AI-Augmented Extended Reality

Xiaoxuan Zhao ^{✉*}Mengting Lai [†]Wen Li [‡]Shiyun Li [§]Pasquale Cascarano [¶]Gustavo Marfia ^{||}

University of Bologna
Department of the Arts
Bologna, Italy

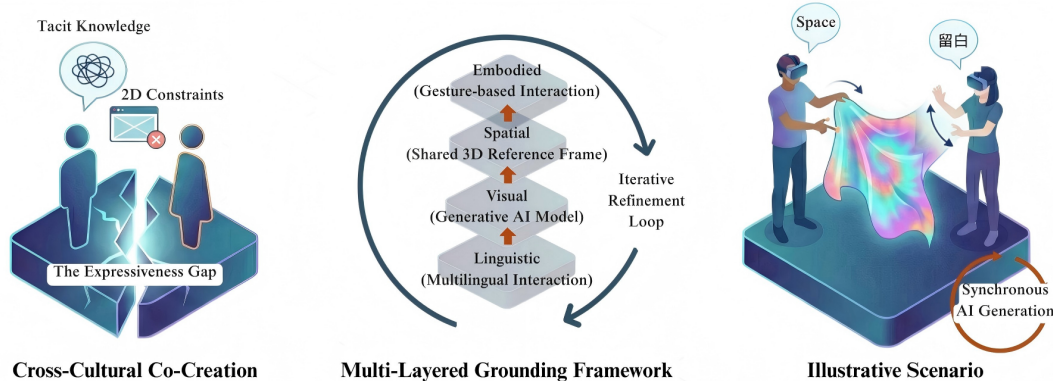


Figure 1: The Multi-layered Grounding Framework bridges the expressiveness gap in cross-cultural co-creation by integrating linguistic, visual, spatial, and embodied grounding in AI-augmented XR environments.

ABSTRACT

A fundamental challenge in cross-cultural co-creation is achieving shared understanding of tacit knowledge when visual and embodied expertise defies verbal description. Our previous work, CrossProbe, demonstrated that Artificial Intelligence (AI)-assisted image generation enables non-expert participants to express concepts visually through natural language, achieving significantly higher cross-cultural understanding and satisfaction scores compared to manual interviews and online questionnaires in a comparative study ($N = 30$). Yet post-study analysis revealed critical limitations—2D representations cannot convey spatial relationships, material qualities, or embodied knowledge. This position paper proposes extending CrossProbe’s mechanisms into eXtended Reality (XR) environments through a multi-layered grounding framework that integrates linguistic flexibility, visual generation, spatial grounding, and embodied interaction. We articulate design principles to ensure inclusive and accessible implementation, democratizing spatial authoring for non-expert participants. This integration addresses expressiveness gaps that neither XR nor AI can solve independently, enabling inclusive cross-cultural collaboration in social XR environments.

Index Terms: Cross-cultural co-creation, Social XR, AI-

*e-mail: xiaoxuan.zhao2@unibo.it

†e-mail: mengting.lai@unibo.it

‡e-mail: wen.li7@unibo.it

§e-mail: shiyun.li3@unibo.it

¶e-mail: pasquale.cascarano2@unibo.it

||e-mail: gustavo.marfia@unibo.it

augmented collaboration, Embodied interaction, Spatial grounding

1 INTRODUCTION

Cross-cultural co-creation faces a fundamental challenge: participants struggle to achieve shared understanding of tacit knowledge when (1) communication relies predominantly on verbal exchange, leaving visual and embodied knowledge unexpressed [6], and (2) common visualization tools such as storyboards and concept maps require significant technical expertise, creating barriers for non-expert participants [7, 18]. Linguistic barriers and cultural conventions exacerbate these difficulties [11]. We term this the “expressiveness gap”—the misalignment between internal understanding and communicative capability [2].

Our previous work, CrossProbe [19], enables participants to generate images in real-time through natural language prompts in their native languages (powered by DALL-E [10]). We developed this tool with two objectives: first, to enable visual communication with greater flexibility than static reference images; second, to make visual expression accessible regardless of artistic or technical expertise. A comparative study ($N = 30$) demonstrated that AI-assisted visual generation significantly improves cross-cultural understanding and participant satisfaction compared to traditional manual interviews and online questionnaires. By externalizing mental models into tangible visual artifacts, CrossProbe effectively reduces the expressiveness gap and lowers participation barriers. However, post-study analysis revealed critical limitations of this 2D approach: static images lack spatial depth cues, fail to convey volumetric properties or material textures, and cannot capture the rich embodied knowledge expressed through physical gestures or spatial orientation.

To more fully bridge the expressiveness gap, this position paper proposes a *multi-layered grounding framework* that integrates linguistic flexibility, visual generation, spatial grounding, and em-

bodied interaction within XR environments. We articulate design principles to democratize spatial authoring, allowing non-expert stakeholders to participate through natural language and intuitive embodied interaction rather than mastering complex technical tools. We argue that spatial computing must enable inclusive cross-cultural collaboration from the outset—ensuring that diverse cultural knowledge systems, including tacit and embodied expertise, can be expressed and preserved on equal terms rather than being filtered through designer-mediated processes.

2 RELATED WORK

We review CrossProbe’s validated mechanisms and three converging research streams that motivate our integrated approach.

2.1 CrossProbe: AI-Assisted Cross-Cultural Chatbot

Our previous work, CrossProbe [19], is a chatbot based on fine-tuned GPT-4 that addresses cross-cultural communication barriers through culturally adaptive dialogue and real-time image generation via DALL-E. Unlike traditional methods requiring high translation costs and technical expertise, CrossProbe enables participants to generate culturally resonant visuals through natural language in their native languages, democratizing visual expression in design collaboration.

A comparative study ($N = 30$) validated its effectiveness across three cross-cultural design research methods. Participants—including native English and Chinese speakers—were assigned to one of three groups: the *chatbot group* used CrossProbe, the *manual interview group* participated in face-to-face interviews, and the *online questionnaire group* completed structured digital surveys. As shown in Tab. 1, the CrossProbe group achieved superior cross-cultural understanding scores and satisfaction ratings.

Qualitative feedback revealed two key mechanisms: (1) the use of native language for image generation increased active participation, and (2) AI-generated visuals captured cultural nuances that generic reference images missed. However, post-study interviews revealed a critical limitation: 2D representations fail to convey spatial relationships, volumetric properties, material qualities, and embodied knowledge. This gap motivates our XR extension with embodied interaction.

2.2 Converging Technologies for Cross-Cultural Co-Creation

Addressing the expressiveness gap requires converging three research streams that have largely developed independently.

Spatial collaboration in XR provides shared coordinate systems where “*this size*” becomes directly perceivable rather than verbally ambiguous [8, 18]. However, requiring 3D modeling proficiency excludes practitioners—from craft artisans to indigenous knowledge holders—whose expertise resides in physical demonstration rather than digital tools [1].

AI-assisted creativity has enabled non-experts to create visual content through natural language [17]. Text-to-2D/3D systems like DALL-E, Hunyuan3D [20] and Meshy¹ have lowered authoring thresholds. However, these systems often produce static outputs for individual viewing and lack mechanisms for real-time, synchronous manipulation in shared environments [12].

Cross-cultural HCI research identifies communication challenges such as linguistic barriers and implicit biases [15]. Current solutions often emphasize human intermediaries [14], which may inadvertently filter non-expert voices.

Our framework addresses these gaps by integrating XR’s spatial grounding, AI’s low-barrier authoring, and cross-cultural inclusivity principles. This integration extends CrossProbe’s validated mechanisms into spatial dimensions, enabling participants

¹<https://www.meshy.ai>



Figure 2: AI-generated assets from natural language descriptions (via DALL-E). This illustrates how linguistic flexibility enables non-experts to express abstract cultural concepts visually in 3D-aware contexts.

to demonstrate techniques through gesture while AI generates spatially manipulable content from multilingual descriptions.

3 POSITION: WHY INTEGRATING XR AND AI IS NECESSARY?

We argue that addressing the expressiveness gap in cross-cultural co-creation requires more than incremental improvements to existing tools. Neither AI-assisted generation nor XR collaboration alone can bridge this gap; their integration is necessary for three primary reasons.

Spatial grounding requires shared coordinate systems. In CrossProbe, when a participant referred to “*this size*,” they gestured in physical space—but this gesture was only visible in the manual interview group, remaining invisible to others. XR solves this through shared spatial coordinate systems: when participants inhabit the same virtual environment, “*this size*” becomes directly perceivable by all, eliminating ambiguity. Critically, this changes the mode of communication. Participants can *reason-through-doing*: arranging virtual fabric samples spatially, comparing proportions directly, and demonstrating draping through embodied movement. This aligns with how tacit knowledge is actually held—in skilled movement and spatial judgment rather than verbal description.

AI bridges technical barriers in XR authoring. A core critique of XR collaboration tools is that they favor technically skilled users, recreating power asymmetries where XR-fluent users gatekeep spatial authoring. AI-assisted generation addresses this by lowering the authoring threshold. Rather than requiring artisans to learn complex 3D modeling software, they describe concepts in natural language—for example, “*A blanket with the rhythmic feel of an ink painting is draped over the chair*” (有水墨画韵律感的毛毯搭在椅子上)—and AI generates visual assets (Fig. 2) directly in the shared space. This preserves CrossProbe’s linguistic flexibility while extending it to spatial domains. Importantly, generation happens synchronously, facilitating collective sense-making.

Embodied interaction externalizes tacit knowledge. The most compelling argument for XR integration comes from CrossProbe’s limitation: participants possess knowledge they can demonstrate but not describe. In XR, participants can “*show how fabric drapes*” through gestures, indicate “*this scale*” relative to virtual avatars, or express “*asymmetrical balance*” through arrangement. This embodied demonstration has no equivalent in 2D media and is essential for preserving the integrity of diverse cultural practices.

Table 1: Comparative analysis of three cross-cultural design research methods from our previous CrossProbe study [19].

KPIs (5-point scale)	Online Questionnaire Group	Chatbot Group	Manual Interview Group	Average Value
Cross-cultural Score	4.3	4.8	4.4	4.5
Satisfaction Score	3.7	4.3	4.1	4.03



Figure 3: Impact of linguistic and cultural nuance on AI-generated imagery. (a) Prompt “留白” (*liúbái*, negative space): The original Chinese term evokes traditional ink wash aesthetics, generating imagery with cultural depth and contemplative negative space. (b) Prompt “empty space”: The literal English translation produces generic botanical illustration lacking artistic intention. Both images generated using DALL-E to demonstrate the necessity of preserving original language inputs in multilingual AI systems.

4 PROPOSED FRAMEWORK: MULTI-LAYERED GROUNDING IN XR

Building on CrossProbe’s validated mechanisms and the three necessity arguments above, we propose a framework integrating four complementary dimensions.

This framework specifically addresses expressive grounding—enabling participants to externalize tacit knowledge and achieve mutual understanding across linguistic, visual, spatial, and embodied dimensions. It does not directly address explicit cultural conflicts, value negotiation when participants hold contradictory design philosophies, institutional power asymmetries, or long-term governance of collaborative processes. These aspects require complementary frameworks from conflict resolution and organizational studies.

4.1 Grounding Layers

Our framework extends Clark and Brennan’s theory [3] of grounding in communication to encompass spatial and embodied dimensions.

Linguistic grounding enables participants to express concepts in their native languages through voice or text input. The framework implements a dual-path approach: translations facilitate mutual comprehension among collaborators (displaying real-time subtitles in shared XR space via multilingual large language models), while generation requests route original language inputs directly to AI models. For instance, when a Chinese participant describes “留白” (*liúbái*), the generative model receives the original term rather than a literal translation like “empty space.” As illustrated in Fig. 3, preserving the original linguistic input allows the AI to capture the specific aesthetic and philosophical intentions encoded in Chinese cultural traditions, which are often lost in translation. This can be implemented through existing multilingual APIs (GPT, translation services) with conditional routing logic that distinguishes between communication and generation contexts.

Visual grounding generates diverse visual representations through text-to-image and text-to-3D models. The framework supports 2D images for conceptualization (patterns, textures spawned as floating planes in XR space), 3D objects and environments for spatial manipulation (furniture, fabrics as physics-enabled meshes), and videos for demonstrating techniques (projected onto virtual screens). Unlike CrossProbe’s 2D output, 3D assets become spatially manipulable within the shared environment—participants can grab, rotate, and scale generated objects through XR controllers or hand tracking. To address generation latency (current text-to-3D models require 10–30 seconds [9]), we propose progressive refinement where low-fidelity previews enable immediate interaction while higher-quality versions generate in background, maintaining conversational fluidity validated in CrossProbe.

Spatial grounding establishes shared coordinate systems through networked XR environments. All participants inhabit the same virtual space with common origin points, ensuring that when one participant places a pattern “here” on a virtual chair, all collaborators see identical spatial relationships. This is implemented through real-time synchronization frameworks (Photon, Mirror [13]) that replicate object transforms, materials, and states across clients. Spatial anchors provide persistent placement, while virtual human figures at 1:1 scale offer universal size reference. Participants can directly manipulate spatial arrangements—comparing proportions through juxtaposition, adjusting scale through gesture—transforming abstract verbal descriptions into concrete spatial judgments.

Embodied grounding captures tacit knowledge through hand tracking and physics-based interaction. XR headsets with hand tracking capabilities (Quest Pro, HoloLens 2) detect finger-level gestures at 60Hz, transmitting hand poses to other participants [5]. For instance, when an artisan demonstrates fabric gathering, their grasping and twisting motions deform virtual textiles with physics simulation (cloth dynamics engines), visible to all collaborators as embodied actions. Full-body avatars with inverse kinematics from head and hand tracking convey posture and body language, providing social presence. The framework can record gesture sequences as replayable animations, preserving embodied techniques for iteration or documentation.

4.2 Layer Integration and Interaction Flow

These four layers operate as an integrated cycle where each dimension enables and amplifies the others. Consider a tie-dye design collaboration:

An artisan initiates with linguistic input: “我想要有水墨画渐变效果的扎染” (*I want tie-dye with ink wash gradient effect*). The system displays the English translation for the remote designer while routing the original Chinese prompt to the generative model (linguistic grounding), generating a pattern image within seconds (visual grounding). The designer then requests, “Can we see this on fabric?” triggering text-to-3D generation of a 1m × 1m cloth mesh with the pattern as a texture, spawned in the center of the shared virtual room (spatial grounding).

The artisan demonstrates gathering techniques by reaching out with tracked hands, pinching the virtual fabric, and twisting—the cloth deforms following physics simulation, visible to the designer in real-time (embodied grounding). Observing the result, the designer suggests “the gradient should be softer” (linguistic iteration), prompting regeneration with modified parameters that updates the

texture on the existing spatial fabric without resetting its position or the artisan's gesture demonstration.

This cycle continues: linguistic descriptions trigger visual generation of spatial content that participants refine through embodied interaction, prompting new linguistic descriptions. Critically, each layer addresses limitations of the others—when verbal description fails, visual generation provides reference; when 2D images lack clarity, spatial manipulation enables exploration; when spatial relationships remain ambiguous, embodied demonstration externalizes tacit knowledge. The integration is synergistic: spatial grounding makes embodied interaction meaningful, visual generation populates the environment with culturally resonant content, and linguistic flexibility ensures all participants can contribute regardless of technical expertise.

4.3 Design Principles for Inclusive Implementation

Finally, we propose three design principles to guide inclusive and accessible implementation.

AI as facilitator, not authority. Generated content must be provisional. Participants can regenerate, modify, or discard outputs. The system should never present AI suggestions as correct designs. Critically, AI assistance must ensure participant parity—all collaborators should have equal capacity to invoke generation regardless of technical fluency or linguistic background, preserving CrossProbe's mechanism of inclusive agency.

Progressive complexity disclosure. Initial interactions use only natural language and gesture. Advanced users may access finer controls (adjust material properties, modify geometry), but baseline functionality remains accessible to non-experts. This mirrors mobile app design: simple core interactions with optional advanced features.

Cultural neutrality in spatial paradigms. The framework must not privilege specific cultural interaction norms. For instance, Western collaboration often assumes forward-facing arrangements (everyone faces the presentation), while many East Asian contexts prefer circular arrangements emphasizing group harmony. The system should provide spatial tools that participants appropriate according to their cultural practices, rather than imposing a single paradigm.

5 DISCUSSION: CHALLENGES AND FUTURE DIRECTIONS

We acknowledge substantial obstacles to realizing this vision. Below we outline implementation challenges and validation pathways for future research.

5.1 Implementation Challenges

Addressing these challenges requires iterative refinement informed by empirical validation.

Technical feasibility: latency versus fluidity. Text-to-3D generation (10–30 seconds [9]) is significantly slower than DALL-E (3–10 seconds) that enabled CrossProbe's conversational rhythm. Multi-user synchronization further compounds coordination overhead. We propose progressive refinement architectures: low-fidelity previews within 3–5 seconds enable immediate manipulation while high-fidelity versions generate in background. Preliminary estimates suggest fluidity degrades beyond 10-second latency, requiring empirical validation of quality-speed-fidelity trade-offs.

Cultural bias: whose aesthetics do models learn? Training datasets skew toward Western content. Testing DALL-E 3 with “留白” (*liúbái*, negative space) produced images lacking the contemplative quality central to Chinese aesthetics (Fig. 3). We propose curating culturally diverse 3D datasets through partnerships with cultural institutions and participatory fine-tuning where communities validate outputs. Critical questions remain: How do we avoid stereotypical representations while enhancing cultural awareness across traditions?

Accessibility: XR hardware as barrier. High-end VR headsets (\$500–3500), motion sickness (20–40% of users), and physical space requirements contradict inclusivity goals [4, 16]. We propose hybrid participation modes: XR users collaborate with 2D interface users through spatial audio, shared annotations, and AI-mediated perspective translation. Mobile AR offers transitional technology with lower barriers. Core functionality should work across platforms through progressive enhancement, with XR providing augmented but not exclusive capabilities.

5.2 Validation Roadmap

We outline validation approaches for each grounding layer to ensure operability for future research.

Linguistic Grounding: Validate whether dual-path processing (original language for generation, LLM translation for communication) preserves cultural semantics better than single-translation approaches. Key metrics include cultural appropriateness of generated assets, cross-language comprehension accuracy, dialogue efficiency.

Visual Grounding: Determine latency thresholds and fidelity requirements for progressive refinement. Key questions include at what delay (3s/5s/10s/15s) does generation disrupt conversational flow? What preview quality enables meaningful spatial reasoning? Does progressive generation reduce total task time?

Spatial Grounding: Quantify disambiguation of spatial deixis in shared 3D reference frames. Measure clarification frequency for expressions like “*this size*” in XR vs. 2D video, variance in spatial interpretations, and accuracy improvement for arrangement tasks.

Embodied Grounding: Assess gesture fidelity and knowledge transfer efficacy. Compare technique replication accuracy (e.g., fabric manipulation) and comprehension between gesture-augmented vs. verbal-only instruction. Evaluate whether physics simulation supports design decisions.

Cross-Layer Integration: Beyond individual layers, validate synergistic effects through comparative studies pairing the full framework against reduced configurations (XR without AI, AI without XR), isolating the integration necessity argued in Section 3.

6 CONCLUSION

As XR becomes integral to collaboration, ensuring it enables rather than obstructs diverse participation is essential. Bridging the expressiveness gap in cross-cultural co-creation requires more than better visual tools—it demands spatial and embodied grounding that enables reasoning-through-doing. We propose a multi-layered framework integrating AI generation with shared XR environments, where linguistic, visual, spatial, and embodied expression work in concert to support inclusive co-creation.

Substantial challenges remain—technical complexity, cultural bias, accessibility barriers and ethical considerations. Yet the potential for more equitable collaboration across cultural boundaries justifies sustained research effort. We present this framework to spark discussion within the XR community, inviting refinement and collaborative development toward systems that enable diverse voices in spatial co-creation.

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